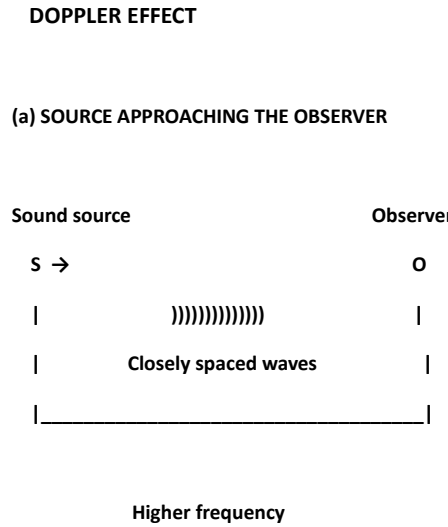
Doppler Effect Simulator

The Doppler effect is the apparent change in the frequency of sound due to relative motion between the source of sound and the observer.



Key observations

- Source approaches the listener: Observed frequency increases; the sound sounds higher-pitched.
- Source moves away from the listener: Observed frequency decreases; the sound sounds lower-pitched.
- Source and listener remain stationary: The observed frequency equals the emitted frequency.

Important formula

For a stationary listener and a moving source:

$f' = f \times v/(v - v_s)$ (source approaching)

$f' = f \times v/(v + v_s)$ (source moving away)

Where:

- f’ = observed frequency
- f = actual frequency of the source
- v = speed of sound in air
- v_s = speed of the source

Remember: The sound itself does not need to change its emitted frequency; the apparent frequency changes because of relative motion.